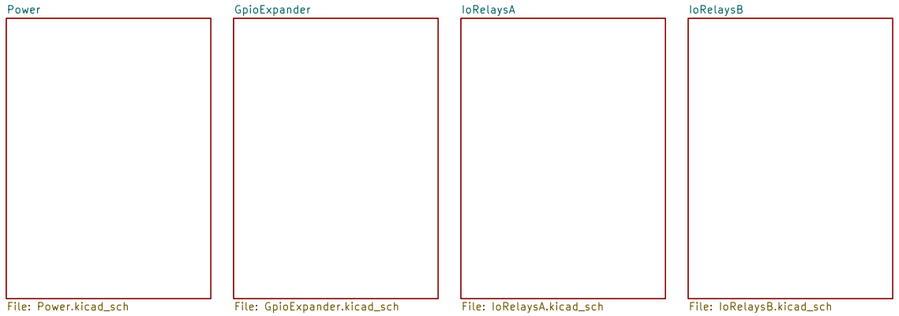
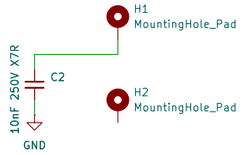




H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
H5 MountingHole_Pad
H6 MountingHole_Pad

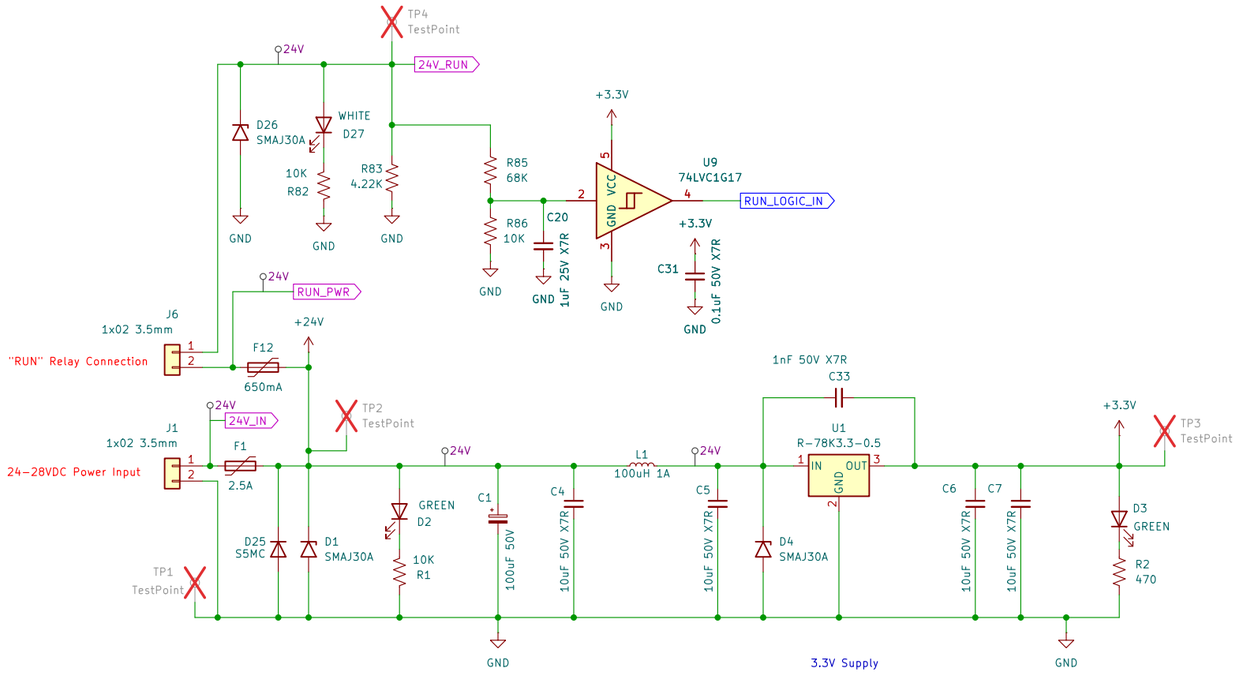
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial

FarmHQ

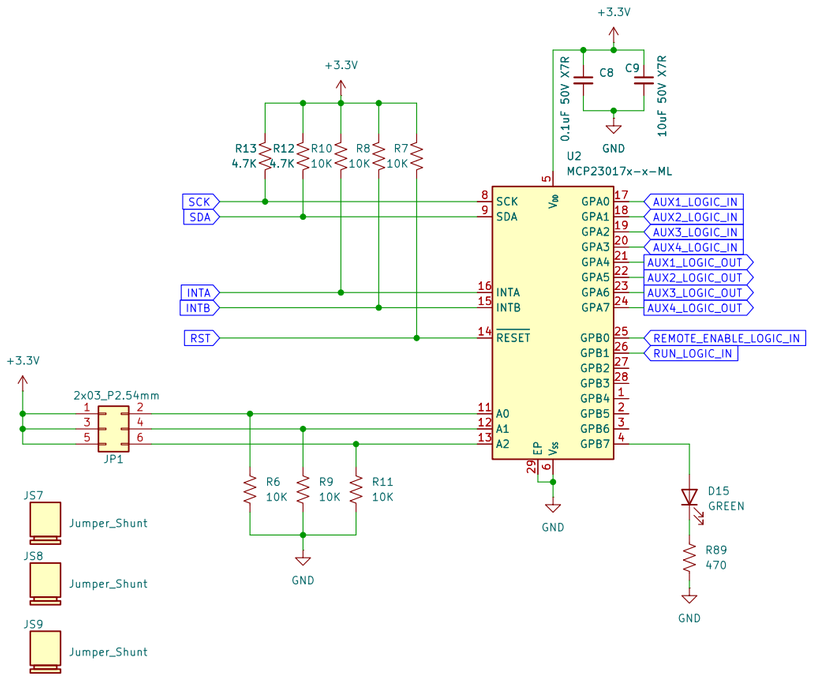
Sheet: /
File: FarmHqRelay_R1.kicad_sch

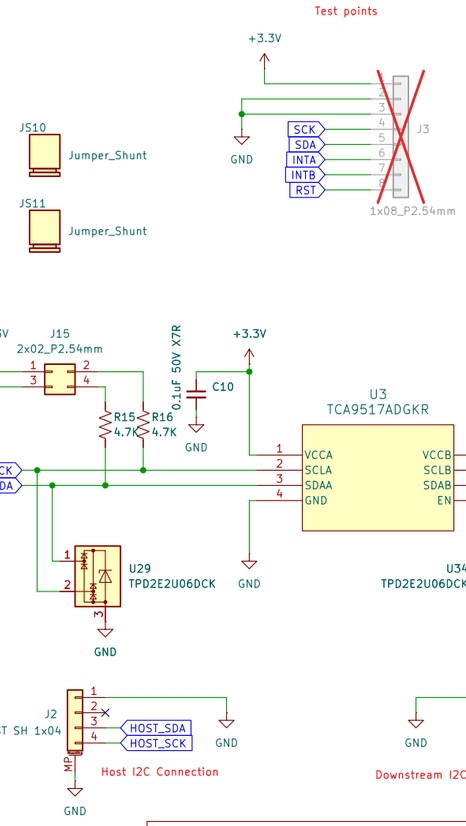
Title: Relay Control Board

Size: USLedger	Date: 2026-10-27	**Rev: 1A**
KiCad E.D.A. 10.0.6		Id: 1/5



FarmHQ	
Sheet: /Power/	
File: Power.kicad_sch	
Title: Relay Control Board	
Size: USLedger | Date: 2026-10-27	**Rev: 1A**
KiCad E.D.A. 10.0.6	Id: 2/5



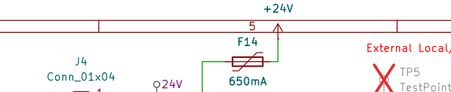


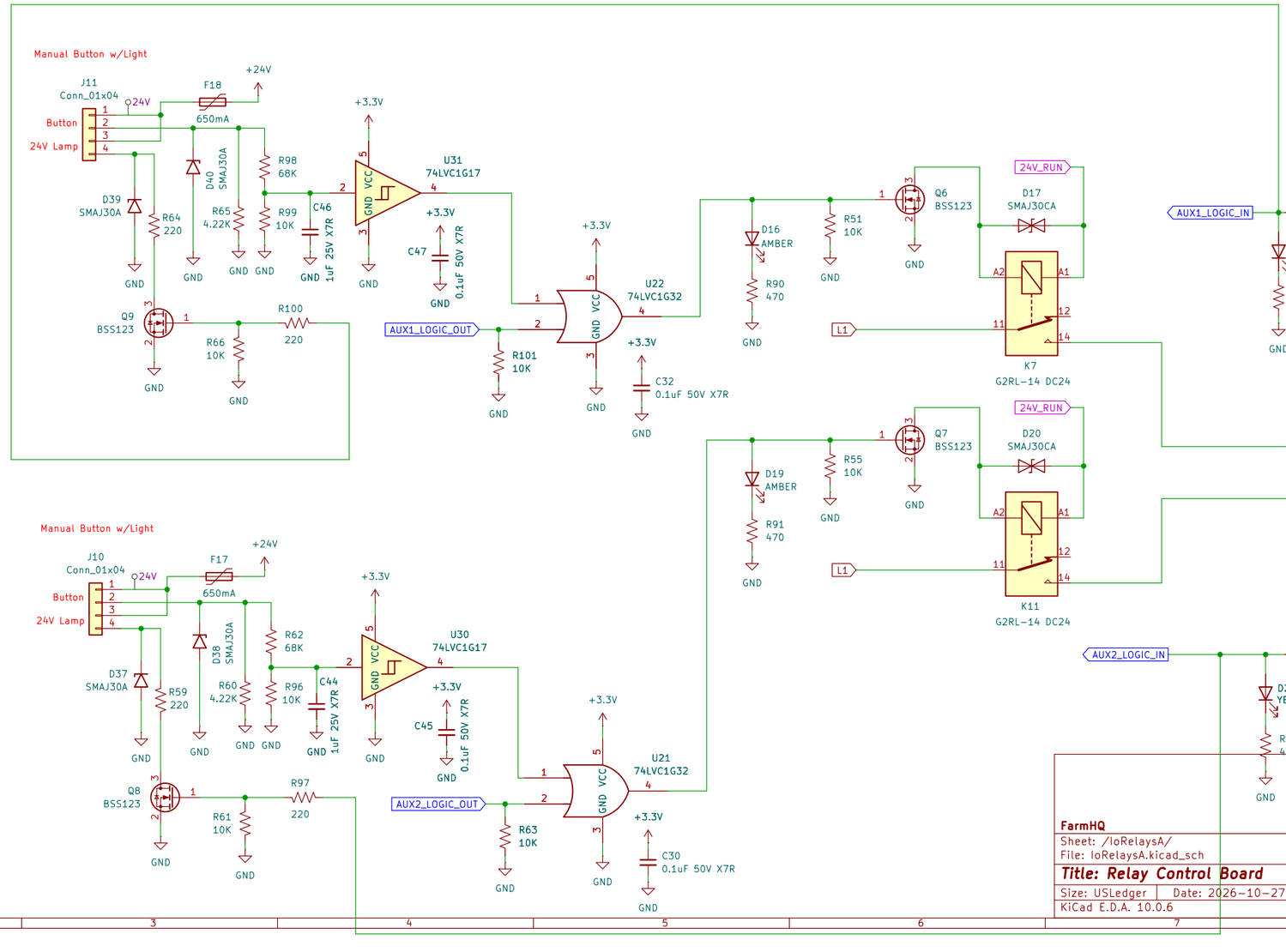
FarmHQ

Sheet: /GpioExpander/
File: GpioExpander.kicad_sch

Title: Relay Control Board

Size: USLedger	Date: 2026-10-27	**Rev: 1A**
KiCad E.D.A. 10.0.6		Id: 3/5




Manual Button w/Light
J11
Conn_01x04
24V
Button
24V Lamp
F18
650mA
+24V
D39
SMAJ30A
R64
220
D40
SMAJ30A
R65
4.22K
R98
68K
R99
10K
C46
1uF 25V X7R
+3.3V
U31
74LVC1G17
GND
VCC
C47
0.1uF 50V X7R
Q9
BSS123
R66
10K
R100
220
AUX1_LOGIC_OUT
R101
10K
U22
74LVC1G32
C32
0.1uF 50V X7R
D16
AMBER
R90
470
R51
10K
Q6
BSS123
24V_RUN
D17
SMAJ30CA
K7
G2RL-14 DC24
L1
Q7
BSS123
R55
10K
D19
AMBER
R91
470
D20
SMAJ30CA
K11
G2RL-14 DC24
Manual Button w/Light
J10
Conn_01x04
F17
650mA
D37
SMAJ30A
R59
220
D38
SMAJ30A
R60
4.22K
R62
68K
R96
10K
C44
1uF 25V X7R
U30
74LVC1G17
C45
0.1uF 50V X7R
Q8
BSS123
R61
10K
R97
220
AUX2_LOGIC_OUT
R63
10K
U21
74LVC1G32
C30
0.1uF 50V X7R
MAINS_BUS
F13
5x20mm Fuseholder
MAINS
NEUTRAL
AUX1_LOGIC_IN
U19
74LVC1G17
C43
0.1uF 50V X7R
D24
YELLOW
R95
470
C29
1uF 25V X7R
R56
22K
U20
FOD814ASD
D18
SMAJ30CA
REMOTE_ENABLE
AUX1_IN
AUX2_IN
F8
F9
K10
G2RL-24 DC24
AUX2_LOGIC_IN
U17
74LVC1G17
C42
0.1uF 50V X7R
D23
YELLOW
R94
470
C28
1uF 25V X7R
R52
22K
U18
FOD814ASD
FarmHQ
Sheet: /IoRelaysA/
File: IoRelaysA.kicad_sch
Title: Relay Control Board
Size: USLedger
Date: 2026-10-27
Rev: 1A
KiCad E.D.A. 10.0.6
Id: 4/5

Manual Button w/Light

J12
Conn_01x04
Button
24V Lamp
24V
F19
650mA
+24V
D41
SMAJ30A
R67
220
GND
D43
SMAJ30A
R69
4.22K
GND
R102
68K
R103
10K
C48
1uF 25V X7R
GND
+3.3V
U32
74LVC1G17
GND
VCC
C50
0.1uF 50V X7R
Q10
BSS123
R70
10K
R105
220
AUX3_LOGIC_OUT
U23
74LVC1G32
R107
10K
C34
0.1uF 50V X7R
D21
AMBER
R92
470
R75
10K
Q12
BSS123
24V_RUN
D28
SMAJ30CA
K8
G2RL-14 DC24
L1
D22
AMBER
R93
470
R76
10K
Q13
BSS123
D29
SMAJ30CA
K12
G2RL-14 DC24

Manual Button w/Light

J13
Conn_01x04
Button
24V Lamp
F20
650mA
D42
SMAJ30A
R68
220
D44
SMAJ30A
R71
4.22K
R73
68K
R104
10K
C49
1uF 25V X7R
U33
74LVC1G17
C51
0.1uF 50V X7R
Q11
BSS123
R72
10K
R106
220
AUX4_LOGIC_OUT
U24
74LVC1G32
R74
10K
C35
0.1uF 50V X7R

AUX3_LOGIC_IN
U25
74LVC1G17
C52
0.1uF 50V X7R
D31
YELLOW
R109
470
C36
1uF 25V X7R
R77
22K
U27
FOD814ASD
D32
SMAJ30CA
REMOTE_ENABLE
AUX3_IN
AUX4_IN
MAINS
K13
G2RL-24 DC24
5x20mm F

AUX4_LOGIC_IN
U26
74LVC1G17
C53
0.1uF 50V X7R
D30
YELLOW
R108
470
C37
1uF 25V X7R
R78
22K
U28
FOD814ASD
NEUTR

FarmHQ
Sheet: /IoRelaysB/
File: IoRelaysB.kicad_sch
Title: Relay Control Board
Size: USLedger
Date: 2026-10-27
KiCad E.D.A. 10.0.6
Rev: 1A
Id: 5/5